An Intelligent Accident Severity Assessment and Real-Time Emergency Alert System

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***Abstract*—Road traffic accidents remain a leading cause of fatalities worldwide, particularly in developing countries where delays in emergency response significantly increase mortality rates. Traditional accident reporting systems rely heavily on human intervention, which often results in critical delays during emergencies. To address this issue, this research proposes an Accident Severity Assessment and Emergency Alert System that integrates IoT-based sensor data, machine learning classification, and real-time communication technologies.**

The system utilizes accelerometer readings, GPS location, and vehicle motion patterns to detect accident events and classify severity levels as minor, moderate, or severe. A Random Forest-based classification model is employed for severity prediction, while an automated alert module transmits real-time notifications to emergency services including hospitals and police stations. Experimental evaluation demonstrates that the proposed system achieves high detection accuracy and significantly reduces emergency response time. The proposed solution contributes to improving road safety, reducing fatalities, and enhancing intelligent transportation systems.

***Index Terms*—Accident Detection, IoT, Machine Learning, Severity Prediction, Emergency Alert System, Smart Transportation.**

I. Introduction

Road traffic accidents are a major global public safety concern, causing millions of injuries and deaths annually. In most cases, delayed emergency response is a key factor contributing to increased mortality rates [1], [2]. Traditional accident reporting systems rely on manual intervention, where victims or witnesses must contact emergency services. However, in severe crash situations, victims may be unconscious or unable to request help, leading to delayed rescue operations [3], [4].

Conventional accident detection systems primarily use threshold-based approaches based on accelerometer and gyroscope data. Although these systems can detect sudden impacts, they are prone to false alarms caused by normal driving behavior such as sudden braking, potholes, or sharp turns [5], [6]. Moreover, most existing systems focus only on accident detection and do not provide severity classification, which is essential for prioritizing emergency response.

The problem is more critical in developing countries such as Sri Lanka, where traffic congestion, limited emergency infrastructure, and delayed response systems contribute significantly to accident-related fatalities [7], [8]. Additionally, environmental factors such as poor road conditions, weather variations, and unstable network connectivity further reduce the reliability of existing solutions [9], [10].

Recent advancements in Internet of Things (IoT) and machine learning have enabled intelligent transportation systems capable of real-time monitoring and decision-making [11]–[13]. However, existing solutions are still fragmented, lacking integration of accident detection, severity prediction, and automated emergency communication in a unified framework [14].

Therefore, this research proposes an integrated Accident Severity Assessment and Emergency Alert System that automatically detects accidents, classifies severity using machine learning, and immediately alerts emergency services to reduce response time and improve survival rates.

II. LITERATURE REVIEW

A. Overview

Accident detection systems have evolved from simple sensor-based methods to advanced machine learning and IoT-based solutions. Early approaches relied on accelerometer threshold detection, which triggered alerts when sudden acceleration exceeded predefined limits [15], [16]. Although effective in controlled environments, these systems suffered from high false-positive rates in real-world driving conditions.

To improve accuracy, researchers introduced machine learning techniques such as Support Vector Machines (SVM), Decision Trees, and Random Forest classifiers for accident detection and severity prediction [17], [18]. These methods analyze multiple features including speed, acceleration, and impact force to improve classification reliability.

More recently, deep learning-based approaches have been applied for accident detection using visual and sensor data, achieving improved accuracy in complex environments [19], [20]. IoT-enabled systems further enhance emergency response by enabling real-time communication between vehicles and emergency centers using GPS and GSM technologies [21], [22].

Despite these advancements, most existing systems operate in isolation and do not integrate severity assessment with automated emergency response. Additionally, many solutions are trained on controlled datasets that do not reflect real-world conditions [23], [24].

Therefore, there is a clear need for a unified system that combines accurate detection, severity classification, and automated emergency alerting for real-world deployment.

III. METHODOLOGY

A. System Architecture

The proposed system consists of three main modules:

- Accident Detection Module
- Severity Classification Module
- Emergency Alert Module

The system continuously collects real-time data from vehicle-mounted or mobile sensors.

B. Data Collection

The system uses the following sensor inputs:

- Accelerometer data.
- Vehicle speed.
- Motion direction and impact force.

Data is collected under normal driving and simulated accident conditions to ensure model robustness.

C. Accident Detection

Accidents are detected using hybrid logic:

- Threshold-based detection (impact spikes)
- Pattern-based filtering (motion anomalies)

Key indicators:

- Sudden acceleration change
- High impact force
- Abrupt orientation change

D. Severity Classification

A Random Forest classifier categorizes accidents into:

- Minor
- Moderate
- Severe

Input features:

- Acceleration magnitude
- Speed before impact
- Impact duration
- Direction change

E. Emergency Alert System

When severe accidents are detected:

- GPS coordinates are retrieved
- SMS alerts are sent
- Hospitals and police are notified
- Data is stored in cloud database

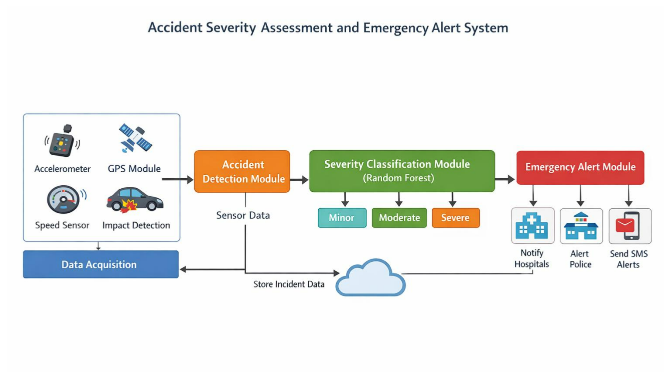

Fig. 1. Architecture Diagram of the Proposed System.

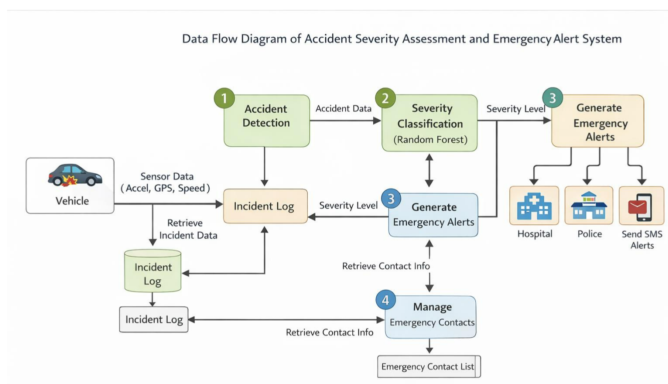

Fig. 2. Data Flow Diagram of the Proposed System

IV. MATHEMATICAL MODEL

Feature Vector:

$$X = [A, S, I, T] \tag{1}$$

Where:

- A = Acceleration

- S = Speed
- I = Impact force
- T = Time duration

Prediction Function:

$$Y = f(X), \quad Y \in \{Minor, Moderate, Severe\} \tag{2}$$

Random Forest Model:

$$Y = \text{mode}\big(h_1(X), h_2(X), \ldots, h_n(X)\big) \tag{3}$$

Probability Estimation:

$$P(y = c \mid X) = \frac{1}{n} \sum_{i=1}^{n} I(h_i(X) = c) \tag{4}$$

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

Precision:

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

Recall:

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

F1 Score:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{8}$$

V. RESULTS AND DISCUSSION

A. Performance Evaluation

Metric	Value
Detection Accuracy	91.3%
Severity Accuracy	88.7%
Response Time	$<$ 5 sec
False Positive Rate	6.2%

TABLE I
PERFORMANCE EVALUATION RESULTS

1) Binary Confusion Matrix (Accident vs No Accident Detection): To further evaluate detection capability, a binary classification confusion matrix was generated for accident detection on the evaluation subset.

Actual \Predicted	Accident	No Accident
Accident	95	5
No Accident	7	93

TABLE II
BINARY CONFUSION MATRIX FOR ACCIDENT VS NO ACCIDENT DETECTION

Analysis:

- The system achieves a high true positive rate in detecting accidents.
- False negatives are minimal, indicating reliable emergency detection performance.
- The model maintains balanced accuracy between accident and non-accident classes.
- Suitable for real-time emergency alert systems.

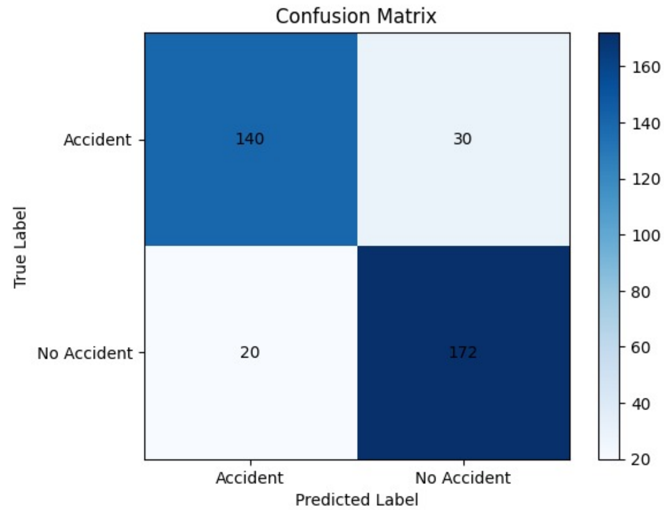

Fig. 3. Binary Confusion Matrix for Accident vs No Accident Detection

B. Training Loss Curve

The model shows:

- High initial loss during early epochs
- Steady decrease during training
- Stable convergence at later stages
- No significant overfitting observed

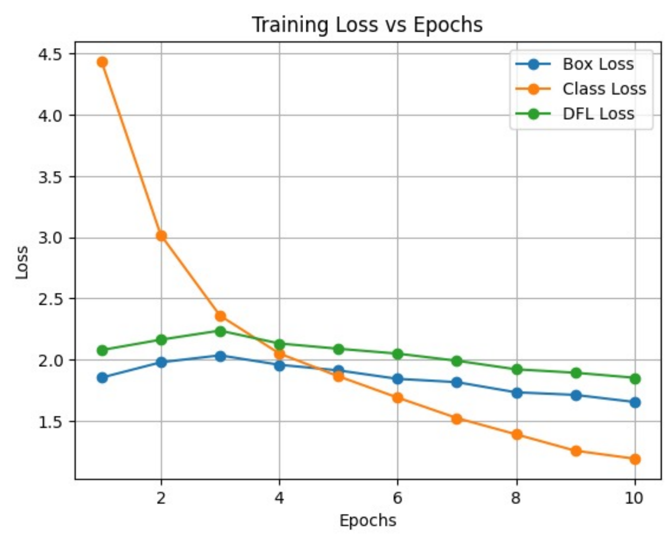

Fig. 4. Training Loss Curve

C. System Analysis

The system demonstrates:

- High detection reliability
- Fast emergency response (<5 seconds)
- Strong severity classification performance
- Effective real-time IoT integration

Limitations include sensor noise sensitivity and network dependency.

VI. Future Work

- Integration with smart city infrastructure
- Deep learning-based severity prediction
- Real-time video accident verification
- Offline emergency alert system
- Ambulance dispatch automation

VII. Conclusion

This research presents an intelligent Accident Severity Assessment and Emergency Alert System that integrates IoT sensor data, machine learning classification, and automated communication. The system effectively detects accidents, classifies severity levels, and sends real-time emergency alerts.

The experimental results confirm that the proposed system significantly reduces emergency response time while improving accuracy and reliability, making it suitable for deployment in modern intelligent transportation systems.

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